

Response to Reviewer 1 — sensors-4464625 (Major Revision)

Manuscript: Occlusion-Aware Visibility Coverage for Robotic Stop-and-Scan 3D LiDAR Mapping

We thank the reviewer for the careful and constructive review and for the assessment that the work shows promise for publication in *Sensors*. Two changes cut across many of the comments: (i) a new experiment — a disk baseline governed by the identical marginal-gain rule as the proposed policy, replayed on the same fixed candidate paths (Sections 4.2 and 6.2; Table 2; Figure 8) — and (ii) explicit 2D scoping of the contribution (new Section 3.5, plus title, abstract, contributions, discussion, and conclusions). In response to the evaluation form, a careful English revision pass has also been made over the entire manuscript (Comment 10). All changes are visible in the marked-up manuscript; page numbers refer to the revised clean manuscript. We follow the numbering of the review report; as it contains two comments numbered 6, we answer them as 6a and 6b.

Comments 1: *The manuscript is strongly positioned within mobile robotics, while the title and terminology are framed primarily around terrestrial laser scanning, which is commonly associated with surveying and geomatics. Please clarify whether the contribution is intended specifically for robotic platforms carrying survey-grade terrestrial laser scanners or for robotic stop-and-scan LiDAR mapping more generally. The terminology should then be applied consistently throughout the manuscript. A possible revised title can be “Occlusion-Aware Visibility Coverage for Robotic Stop-and-Scan 3D LiDAR Mapping”*

Response 1: Thank you for pointing this out. We agree with this comment. Therefore, we have adopted the suggested title verbatim. The contribution is intended for robotic stop-and-scan LiDAR mapping in general; the survey-grade TLS (BLK360 G1) is the canonical instance of that sensor class and our experimental instrument. The revised Introduction now defines the regime explicitly (Section 1, p. 2): “We refer to this regime as *robotic stop-and-scan LiDAR mapping*: a mobile platform carries a light detection and ranging (LiDAR) sensor that must remain stationary during each acquisition, and the dense scans are aligned offline rather than registered incrementally in real time. A terrestrial laser scanner (TLS) is the canonical instance of such a sensor ... The method developed in this paper is formulated for the general stop-and-scan LiDAR regime; the BLK360 G1 enters only as the experimental instrument (Section 5).” Terminology has been made consistent throughout: the method sections (Sections 3–4) now speak of the generic stop-and-scan sensor with range bound R and stationary acquisition cost, and instrument-specific values appear only in the experimental sections (see also our response to Comment 5).

Comments 2: *The research gap is reasonably described, particularly in Section 2.4, where the authors identify the intersection between online exploration of unknown environments and planning for expensive stationary sensing. However, the manuscript does not formulate explicit research questions. The objectives and hypotheses must currently be inferred from the contributions and experimental sections. Please state one principal research question, supported where appropriate by subquestions. The Introduction should also conclude with a short roadmap explaining the structure of the remaining sections.*

Response 2: We agree, and the Introduction now states one principal research ques-

tion and three subquestions (Section 1, p. 3): “**RQ.** *In online robotic exploration of an unknown indoor environment with a stop-and-scan LiDAR sensor, does an occlusion-aware visibility coverage model produce better scan placement — higher genuinely observed coverage at a comparable number of stationary scans — than the conventional proximity-based disk model?*” with subquestions (RQ1) how per-pose coverage should be defined on the live map so it never claims occluded or unmapped space (Section 3); (RQ2) how much of the placement difference is attributable to the coverage model alone, once exploration stochasticity *and the acceptance rule* are controlled for (Sections 4 and 6); and (RQ3) whether the effect carries over to a physical platform and to the registered survey product (Sections 6.6 and 6.7). A structure roadmap paragraph now closes Section 1 (pp. 3-4).

Comments 3: *The proposed visibility model operates on a two-dimensional occupancy grid, whereas the resulting product and registration evaluation concern three-dimensional point clouds. The LOS metric therefore evaluates visible floor-space cells rather than the actual 3D surfaces observed by the scanner. It is unclear how the model accounts for vertical occlusions, shelves, elevated machinery, ceilings, overhanging structures, or surfaces that occupy the same horizontal position at different heights. High 2D LOS coverage does not necessarily imply comprehensive 3D surface coverage. The authors should either: extend the method toward a genuinely 3D visibility representation, such as a voxel-, octree-, or surface-based model; or explicitly define the contribution as 2D scan-station planning for subsequent 3D mapping, discuss this limitation, and identify full 3D visibility reasoning as future work. This distinction should be reflected consistently in the title, abstract, contributions, discussion, and conclusions.*

Response 3: A fair and important point. We have taken the second option, and the 2D scope is now stated explicitly and consistently:

- **New Section 3.5 “Scope: 2D Station Planning for 3D Mapping” (p. 8)** states that the model reasons on a horizontal slice at sensor height; that LOS coverage certifies floor-space sight lines, not 3D surface coverage; that vertical occlusions (shelves, overhangs, elevated machinery, height-stacked surfaces) are not represented; why the 2D scope is matched to the online setting (the occupancy grid is what SLAM provides in real time; the 2D polygon is cheap enough to evaluate at every candidate; a full-dome scanner observes much of the vertical structure from a pose with clear 2D LOS); and that a genuinely 3D visibility representation (voxel-, octree-, or surface-based) is future work.
- **Abstract (p. 1):** “...a ray-cast visibility region computed on the robot’s live two-dimensional (2D) occupancy map ... The planner thus performs 2D scan-station placement for subsequent 3D mapping.”
- **Contributions (p. 3):** the first contribution is now “defined on the live 2D occupancy grid”.
- **Discussion (Section 7, p. 24):** the first limitation is now the 2D representation and vertical occlusion, with 3D visibility identified as the principal future-work direction.
- **Conclusions (Section 8, p. 24):** “...the model performs 2D scan-station planning on the live occupancy map for subsequent 3D mapping”; future work includes the 3D extension.

Comments 5: *The method should be formulated independently of the specific instrument used in the experiments. The Leica BLK360 G1 is an experimental implementation rather than an inherent component of the proposed algorithm. The BLK360 scan duration and accuracy should be presented as experimental settings rather than general characteristics of the proposed method.*

Response 5: Agreed, and done. The method sections no longer reference the instrument: Section 4’s cost motivation now reads “minutes per acquisition for the survey-grade scanner used in our experiments (Section 5)” instead of quoting BLK360 scan times; the coverage-model figure caption (Figure 1, pp. 6–7) and the concept-figure title now refer to a generic scan pose; the summary and sweep figures label “stationary scans” rather than “BLK360 scans”. BLK360 duration, accuracy, and workflow details appear only in Section 5 (experimental setup, pp. 10–12) and in the hardware sections, as experimental settings.

Comments 6a: *Several broad claims require stronger references, particularly the statement that the coverage model using an isotropic disk is central to the scan-placement decision. Relevant literature on next-best-view planning, visibility analysis, view planning, and sensor placement should be cited. The manuscript groups structured-light scanners, thermal panorama systems, ground-penetrating radar, gas sensors, spectral sensors, and NDT probes as “high-precision stationary sensors.” These instruments vary in observation geometry, resolution, penetration, and uncertainty. They are better described as sensors needing stationary acquisition or dwell time. The current line-of-sight model is not directly applicable to penetrating sensors like GPR. In abstract, the statement that occlusion-aware visibility “is the appropriate basis” for stop-and-scan placement is too definitive given the limited environments, 2D representation, selected platform, and single scanner type.*

Response 6a: We have made all three changes:

- *Disk-model convention claim:* now supported with citations where first made (Section 1, p. 2) — “a sensor-footprint approximation widely used to drive on-line decisions in frontier exploration and coverage practice [Yamauchi 1997; Umari & Mukhopadhyay 2017; Choset 2001; Galceran & Carreras 2013; Placed et al. 2023].” The related-work section additionally cites the NBV/view-planning literature (Scott 2003; Tarabanis 1995; Zeng et al. 2020) and the visibility-based scan-planning literature (Mozaffar & Varshosaz 2016; Prieto et al. 2017; Dehbi et al. 2021; Knechtel et al. 2025) (Sections 2.1–2.3, pp. 4–5).
- *Sensor grouping:* the family is now described as “sensors that require stationary acquisition or a dwell time at each viewpoint” (p. 2), and we state explicitly: “These instruments differ widely in observation geometry, resolution, and uncertainty; the line-of-sight coverage model developed here applies to the optical, non-penetrating members of this family and explicitly not to penetrating modalities such as GPR (Section 7).”
- *Abstract claim:* softened and scoped (p. 1) — “The results indicate that, for the room-scale indoor environments studied, occlusion-aware visibility is a sounder basis than Euclidean proximity for stop-and-scan placement.” The Conclusions carry the same scoping (“Within the room-scale indoor environments and the optical stop-and-scan sensor class studied here...”, p. 24).

Comments 6b: *The manuscript states that it extends a conference paper that is*

currently under review. Please verify that this is consistent with the journal's editorial policy and clearly disclose the overlap between the two submissions. The authors should identify which methods, experiments, figures, and results are new in the journal submission. The sentence stating that the mobile platform was absent from the conference version is unnecessary in the scientific narrative and should be removed.

Response 6b: The Introduction now discloses the overlap explicitly and itemizes what is new (Section 1, p. 3): “The conference version contributes the stop-and-scan system integration (frontier exploration coupled to a scan sequencer) and an isotropic-disk scan trigger, demonstrated in simulation. New to this article are: the ray-cast visibility coverage model and the LOS coverage metric (Section 3); the dual-criterion marginal-gain placement rule and the coverage-completion phase (Section 4); the controlled paired ablation with its three baselines and the parameter sensitivity analysis (Section 6); and the entire hardware study — the fully autonomous on-hardware disk-versus-visibility comparison and the per-run offline registration of the acquired scans (Sections 6.6 and 6.7). No figure, table, or quantitative result is shared between the two manuscripts.” The sentence noting the platform’s absence from the conference version has been removed, as requested. Since the original submission, the conference paper has been **accepted for presentation at IC-CAS 2026** (Regular Paper; decision of 31 July 2026); its status has been updated in reference [9], in the Introduction, and in the extended-version footnote on the first page. The accepted conference manuscript has been provided to the editorial office for the editors and reviewers, and we have confirmed that this extension route is consistent with MDPI’s editorial policy on extended conference papers.

Comments 7: *Equations (1)–(7) are generally understandable, but several definitions should be clarified: Define the grid-cell domain, cell-center coordinates, grid resolution, and ordering of the occupancy thresholds. Distinguish clearly between a grid cell and a candidate scan pose; the symbol (c) is probably used for both. Explain whether visibility regions associated with previous scan poses are frozen or recomputed when the online occupancy map changes. Describe Equation (6) as a normalized marginal gain and define it only for a nonempty visibility region. State the square-cell assumption required by Equation (7) and report the numerical grid resolution. The relationship between the disk baseline and Equations (6)–(7) also needs clarification. Skipping a candidate because its center lies within (R) of an existing scan is not mathematically equivalent to applying the marginal-gain rule B_{disk} . If the baseline uses a different acceptance rule, the comparison changes both the coverage model and the decision rule.*

Response 7: All requested clarifications have been made in Sections 3–4:

- **Grid domain/resolution/ordering** (Section 3.1, p. 6): “The grid is an $H \times W$ array of square cells of side ρ , the grid resolution ($\rho = 0.05$ m in all experiments); each cell c is identified with its center point in the map frame, so that $\|c - c'\|$ denotes the Euclidean distance between cell centers. ... thresholds ordered as $0 \leq \theta_{\text{free}} < \theta_{\text{occ}} \leq 100$.”
- **Cell vs. candidate pose** (Section 3.1, p. 6): the candidate pose is now denoted $\mathbf{p}$ throughout Section 4 and Algorithm 1 (“Throughout the paper, s_i denotes an accepted scan pose and $\mathbf{p}$ a candidate pose under evaluation, while c is reserved for grid cells”).
- **Frozen or recomputed** (Section 4.1, p. 8): “The visibility regions of previously

accepted scans are not frozen: at every decision, $C(S)$ is recomputed on the current occupancy map, so earlier regions expand or contract as SLAM refines the map (the final reported coverage is evaluated once more on a single reference map; Section 5).” This matches the implementation.

- **Equation (6)** (p. 8): now introduced as the “*normalized marginal gain*”, “defined only for a nonempty visibility region”, with the degenerate-region cut stated to reject candidates *before* the ratio is evaluated, “so the denominator of the normalized gain never vanishes.”
- **Equation (7)** (pp. 8-9): “where cells are squares of side ρ (the grid resolution, $\rho = 0.05$ m), so each cell contributes area ρ^2 and a is in m^2 .”
- **Disk baseline vs. Equations (6)-(7)**: the reviewer is right, and this observation prompted the main new experiment of the revision. We now state explicitly that the spacing rule is *not* mathematically equivalent to the marginal-gain rule applied to B_{disk} (“a candidate closer than R to an existing scan can still contribute a crescent of new disk area”, Section 4.2, p. 9), and we resolve the resulting confound with a **new disk marginal-gain baseline**: the identical dual criterion of Equation (8) with B_{disk} substituted for B , replayed on the same fixed candidate paths (Sections 4.2 and 6.2; Table 2, p. 16; Figure 8, p. 19). Within that pair, the only difference is the coverage model. Results: single-room $77.7 \pm 3.8\%$ LOS at 2.6 ± 0.5 scans (essentially the spacing rule’s result); multi-room **$59.1 \pm 1.3\%$ LOS** at 2.9 ± 0.3 scans — worse than the spacing rule, because the disk model’s own coverage bookkeeping saturates through partitions and stops scanning while whole rooms remain unobserved (mechanism explained in Section 6.2, pp. 15-16). This isolates the deficiency to the coverage model itself rather than the acceptance rule.

Comments 8: *The general trends in Figure 11 are understandable, but the presentation and explanation need clarification. Please: identify the solid LOS-coverage curves and dashed scan-count curves clearly in the legend; avoid describing scan-count changes in “percentage points”; explain how ($A_{\text{min}}=5$ m²) was identified as the knee point; and include variability measures for the averaged results.*

Response 8: The figure has been regenerated (now Figure 8, p. 19, after the renumbering of Comment 13) and the text rewritten:

- The legend now identifies each curve explicitly: “single-room — LOS coverage (solid, left axis)”, “single-room — scan count (dashed, right axis)”, etc.
- All curves now carry **error bars (per-path standard deviation over the $N = 5 / N = 10$ replay paths)**; the caption states “mean $\pm$ s.d.”
- Scan-count changes are no longer described in “percentage points”: “...changes mean LOS coverage by at most about two points (single-room 86.4→84.3%; multi-room 85.2→84.0%) and mean scan count by less than one scan.”
- The knee identification is now an explicit criterion: “it is the largest threshold whose mean LOS remains within about five points of the most aggressive setting ($A_{\text{min}} = 1$ m²) in both environments (85.4 vs. 90.7%; 84.0 vs. 85.7%) while already halving the scan count (3.6 vs. 7.2 scans in both environments). Beyond the knee the trade turns unfavorable: pushing to $A_{\text{min}} = 15$ m² saves only about one further scan (3.6→2.2 and 3.6→2.8) while coverage continues to erode (85.4→80.8% and 84.0→83.1%).”

Comments 9: *Please distinguish between the geometric measurements produced by the laser scanner and the RGB information supplied by the integrated camera. Unless color is used in visibility evaluation or registration, repeated references to “colorized point clouds” can be reduced.*

Response 9: The Introduction now states once (p. 2): “(The scanner also records panoramic RGB imagery that colorizes the point cloud; the planning and evaluation in this paper concern the geometric measurements only.)” Repeated “colorized” phrases have been removed throughout; apart from that definitional sentence, the word now appears only in the caption of the colorized rendering in Figure 15, where color is genuinely shown.

Comments 10: *The long sentence on page 1, lines 25–29 should be divided into shorter sentences. The English should be carefully revised throughout the manuscript, including the long captions and sentences.*

Response 10: The page-1 sentence has been split (“...is growing. A dense and accurate point cloud is becoming an infrastructural asset: downstream robots localize against it, plan collision-free motion within it, and reason over it.”). We have additionally shortened the longest captions (Figures 2, 3, 7, 8, 11, 13–15; Tables 2–3) and made a careful English pass over the manuscript, which also addresses the “Quality of English Language” item of the review form.

Comments 11: *When reference authors are named in the text, the citation should follow their names directly. For example: “Mozaffar and Varshosaz [6]...”*

Response 11: Fixed (Section 2, p. 5): “Mozaffar and Varshosaz [6] optimize scanner placement to reduce occlusions, and Prieto et al. [5] drive a next-best-scan policy...”. The same convention is applied to the newly added named citations (Dehbi et al., Knechtel et al.).

Comments 12: *Define every abbreviation at first use, including TLS, SLAM, LOS, NBV, and MPPI, even if an abbreviation list is provided later.*

Response 12: All abbreviations are now defined at first use: 3D, TLS, LiDAR, SLAM, GPR (Section 1); LOS (Section 1); NBV (Section 2.1); BIM (Section 2.3); MPPI (Section 5.1). The abbreviations list has been extended accordingly (LiDAR, GPR, BIM added; p. 25).

Comments 13: *Figure 4 is cited before Figures 1–3 are introduced. Figures should be numbered and cited in the order of first appearance.*

Response 13: Fixed. The coverage-model figure (previously Figure 4, cited in Section 3 before Figures 1–3 appeared) has been moved to Section 3.2, where it is first cited; it is now Figure 1 (pp. 6–7), and all figures are numbered in order of first citation.

Comments 14: *The labels in Figure 1 are too small at the normal manuscript scale. Please enlarge the text and simplify the diagram where possible. The caption should also be shortened, with detailed explanations moved into the main text.*

Response 14: The architecture diagram (now Figure 2, p. 11) has been redrawn: labels are roughly twice as large at manuscript scale, the layout is a compact two-row pipeline, the redundant color legend was removed (the containers are labeled directly), and the caption has been shortened to two sentences with the component description kept in the Section 5.1 text.

Comments 15: *The industrial test room and its relevant characteristics are described mainly in the Figure 3 caption. The environment should first be introduced in the main text, followed by a citation to the figure.*

Response 15: The test room is now introduced in the main text (Section 5.2, p. 12): open exhibition/staging space, reflective floor, display wall of monitors and product panels, freestanding equipment cases and furniture creating self-occlusion, floor markings for the survey reference layout, and its approximate extent of 16×8.5 m (about 105 m² of mapped free floor area). The figure caption has been reduced to a single sentence.

Comments 16: *Report millimeter-scale registration errors in millimeters rather than meters. For example: (0.004–0.005 m) should be written as 4–5 mm; (0.003–0.006 m) should be written as 3–6 mm. The units should be consistent throughout the manuscript.*

Response 16: All registration errors are now reported in millimeters (Sections 6.7, pp. 21–22): bundle errors 4–5 mm, per-link errors 3–6 mm, disk single-link 2 mm, quality band ≤ 15 mm. Table 3's column is now "Bundle error (mm)". Units are consistent throughout the text, tables, and abstract.

Comments 17: *Several explanations are repeated in the abstract, introduction, results, discussion, figure captions, and conclusions, particularly those concerning: colorized point clouds; survey-grade registration; single-link and multi-link registration networks; and the difference between disk and visibility coverage. The manuscript would benefit from removing this repetition.*

Response 17: We removed repeated explanations of colorized point clouds (now defined once), the single-link/multi-link registration contrast (explained once in Section 6.7 and only referenced elsewhere), and the disk-vs-visibility difference (defined in Sections 1 and 3 and thereafter referenced); shortened captions no longer duplicate the surrounding text.

Comments 19: *Many figure and table captions are unnecessarily long. Please shorten them and move interpretation or methodological detail into the main text.*

Response 19: Captions of Figures 2, 3, 7, 8, 11, 13, 14, 15 and Tables 2, 3 have been shortened, with interpretation and methodological detail moved into the main text (Sections 5.1, 5.2, 6.2, 6.5–6.7).

Response to Reviewer 2 — sensors-4464625 (Major Revision)

Manuscript: Occlusion-Aware Visibility Coverage for Robotic Stop-and-Scan 3D LiDAR Mapping

Manuscript: Occlusion-Aware Visibility Coverage for Robotic Stop-and-Scan 3D LiDAR Mapping We thank the reviewer for the constructive review. The most substantial change prompted by this review is a **new experiment**: a disk baseline governed by the *identical* accept/skip rule as the proposed policy (see Comments 2 and 5 below). Each comment is quoted before its response; all changes are visible in the marked-up manuscript.

Comments 1: Overall, the paper is well presented...

Response 1: We thank the reviewer for the positive overall assessment and address each of the specific comments below in turn.

Comments 2: It is not clear why the isotropic disk method alone is sufficient to evaluate the proposed approach. Is this baseline representative of the current state of the art? Why additional methods are not considered in the comparison? Additional comparisons, or a stronger justification for the selected baseline it is necessary to strengthen the experimental evaluation.

Response 2: We have strengthened the evaluation in both of the ways the reviewer suggests.

Additional comparison. The revision adds a fourth policy to the controlled ablation: a **disk marginal-gain baseline** that applies the identical accept/skip rule (Equation (8)) as the proposed policy, with the disk coverage model B_{disk} substituted for the visibility region B . The four compared policies are now: (i) uniform no-skip (the fixed-interval policy of stop-and-go practice; upper reference), (ii) disk spacing (proximity rule), (iii) disk marginal-gain (matched rule; **new**), and (iv) ray-cast visibility (ours). See Sections 4.2 and 6.2, Table 2, and Figure 8. Notably, the matched-rule disk baseline reaches only **59.1 ± 1.3% LOS** in the multi-room world — worse than even the spacing rule — because the disk model’s through-wall overclaiming saturates its own coverage bookkeeping (mechanism explained in Section 6.2).

Justification of the baseline set. Section 6’s preamble now justifies this set explicitly: the disk/proximity treatment is a commonly used online coverage model in stop-and-go practice (Section 2), while published TLS scan-planning methods — including the strongest recent ones, now cited (Dehbi et al., ISPRS J. Photogramm. Remote Sens. 2021; Knechtel et al., ISPRS J. Photogramm. Remote Sens. 2025) — optimize viewpoints *offline over a prior environment model* (floor plan, BIM, or coarse scan) and are therefore not applicable as baselines in the online unknown-environment regime studied here, where every decision must be made on the live SLAM map. NBV planners for continuous sensors likewise do not decide whether to pay a large fixed stationary cost. The uniform no-skip reference bounds the coverage attainable along each path, so the reported gains are located within a bracketed range rather than against a single baseline.

Comments 2 (continuation): Since the present manuscript is explicitly described as an extension of the conference work, the authors should clearly identify the differences and the additional contributions of the present manuscript. Moreover, since

reference 9 is not yet available, the conference manuscript should be made available to the editors and reviewers.

Response 2 (continuation): The Introduction now discloses the overlap item by item:

“The conference version contributes the stop-and-scan system integration (frontier exploration coupled to a scan sequencer) and an isotropic-disk scan trigger, demonstrated in simulation. New to this article are: the ray-cast visibility coverage model and the LOS coverage metric (Section 3); the dual-criterion marginal-gain placement rule and the coverage-completion phase (Section 4); the controlled paired ablation with its three baselines and the parameter sensitivity analysis (Section 6); and the entire hardware study — the fully autonomous on-hardware disk-versus-visibility comparison and the per-run offline registration of the acquired scans (Sections 6.6 and 6.7). No figure, table, or quantitative result is shared between the two manuscripts.”

The conference manuscript has been uploaded to the submission system for the editors and reviewers, as requested. Since the original submission, the conference paper has been **accepted for presentation at ICCAS 2026** (Regular Paper; decision of 31 July 2026), and its status has been updated accordingly in reference [9], in the Introduction, and in the first-page extended-version footnote.

Comments 3: For equation 1, it is not clear why the authors selected the thresholds of 25 and 65.

Response 3: Section 3.1 now explains the choice: these are the long-standing free/occupied convention of the ROS occupancy-map pipeline (probability thresholds 0.25/0.65), and the classification is insensitive to their exact placement because SLAM-estimated occupancies are strongly bimodal — almost all mapped cells settle near 0 or near 100 — so any thresholds separating the two modes yield the same classification. The deliberately separated band maps ambiguous cells to *unknown*, which the visibility model treats conservatively: they block rays and are never counted as covered.

Comments 4: For equation 3, it would be useful to clarify if the proposed equation represents the ideal geometric visibility, since the equation does not explicitly include K (the number of rays) or the angular resolution used in the presented approach.

Response 4: Correct, and this is now stated explicitly after Equation (3): “the set definition above characterizes the ideal geometric visibility region and does not depend on the ray discretization; the K rays and half-cell marching are the implementation that approximates it. At the range bound the inter-ray arc spacing is $R \cdot \Delta\theta \approx 5.2$ cm for $R = 6$ m, on the order of a single grid cell ($\rho = 5$ cm), so the discrete approximation recovers the ideal region up to single-cell effects at region boundaries.”

Comments 5: However, it is not clear if the same acceptance logic is applied to the isotropic disk baseline, since later it is described as skipping candidates located within R of an existing scan. Therefore, the statement “The only difference between rows is the coverage model used for the scan/skip decision” in table 2 should be reconsidered.

Response 5: The reviewer is right, and this observation prompted the main new experiment of the revision. The spacing rule is *not* the marginal-gain rule applied to B_{disk} , and Section 4.2 now says so explicitly (“a candidate closer than R to

an existing scan can still contribute a crescent of new disk area”). To make the comparison well-posed, we added the **disk marginal-gain baseline** described under Comment 2: the identical Equation (8) rule with B_{disk} substituted for B and disk-based bookkeeping of the covered set, replayed on the same fixed candidate paths. Table 2 now contains four rows per environment, and its caption has been corrected to:

“...the disk marginal-gain row applies the identical accept/skip rule as the visibility policy, so within each environment that pair differs only in the coverage model.”

The former claim (“the only difference between rows is the coverage model”) has been removed. The outcome sharpens the paper’s conclusion: under the identical acceptance rule, the coverage models are separated by 24.9 points of LOS in the multi-room world (59.1% vs. 84.0%), attributing the deficiency to the disk coverage model itself rather than to the acceptance rule attached to it.

Comments 6: The references should be updated as the majority of the references were published before 2020. Please update the references to better reflect the current state of the art.

Response 6: We added six recent works and cite them where they bear on the argument: Zeng et al. 2020 (view-planning survey); Schmid et al. 2020 (sampling-based online informative path planning); Zhou et al. 2021 (FUEL, incremental frontier exploration); Dehbi et al. 2021 and Knechtel et al. 2025 (ISPRS J. Photogramm. Remote Sens.; optimal TLS scan planning with network-connectivity constraints, and joint standpoint + routing optimization for stop-and-go scanning with network redundancy). Together with the already-cited 2020+ works (Placed et al. 2023; Lluvia et al. 2021; Park et al. 2024; Aryan et al. 2021; Halder & Afsari 2023; Quin et al. 2021; He et al. 2025; Macenski et al. 2020), 16 of the 39 reference entries are now dated 2020 or later; of these, 13 are peer-reviewed archival publications (2020-2025), the remainder being the scanner datasheet, an open-source software reference, and our own conference paper accepted at ICCAS 2026. The two ISPRS papers anchor the related-work discussion of current TLS scan-planning practice.

Response to Reviewer 3 — sensors-4464625 (Major Revision)

Manuscript: Occlusion-Aware Visibility Coverage for Robotic Stop-and-Scan 3D LiDAR Mapping

Manuscript: Occlusion-Aware Visibility Coverage for Robotic Stop-and-Scan 3D LiDAR Mapping We thank the reviewer for the careful annotation of the manuscript PDF. We identified twelve annotated comments; each is quoted below with the passage it marks, followed by our response. All changes are visible in the marked-up manuscript. (Page numbers refer to the reviewed submission.)

Comments 1 (p. 1): *"this is difficult to decipher"* (on the sentence "...a dense, accurate point cloud is becoming an infrastructural asset that downstream robots localize against, plan collision-free motion within, and reason over.")

Response 1 (p. 1): The sentence has been split and simplified: "...is growing. A dense and accurate point cloud is becoming an infrastructural asset: downstream robots localize against it, plan collision-free motion within it, and reason over it."

Comments 2 (p. 1): *"the question of why isn't answered here — time, safety, less of an interruption to processes?"* (on "...rather than through labor-intensive manual scanning.")

Response 2 (p. 1): Exactly these motivations are now stated in Section 1: "The motivation is threefold: a manual survey occupies a trained operator for hours of tripod placements; a robotized acquisition can run outside operating hours, interrupting neither production processes nor people on the floor; and an autonomous procedure makes periodic re-scans repeatable at low marginal cost."

Comments 3 (p. 2): *"this is an issue with that particular piece of hardware, but not all 3D scanners; this should be clarified because it infers that the problem is universal."* (on "the dense scans are not registered in real time but aligned offline. Conventional mobile mapping...")

Response 3 (p. 2): Agreed. Section 1 now scopes the two defining properties explicitly to the sensor class rather than to 3D scanning at large: "We emphasize that these two properties characterize the survey-grade stop-and-scan class, not 3D laser scanning in general: continuous mobile-mapping scanners acquire under motion and register incrementally online through simultaneous localization and mapping (SLAM), trading survey-grade accuracy for acquisition speed." The related claim in Section 2.3 is scoped the same way ("Survey-grade terrestrial scanning, as practiced with the tripod-class instruments considered here, departs from the continuous-acquisition assumption... (Scanners that relax these constraints exist, but at accuracies below the survey grade targeted here.)"); see also Comment 8.

Comments 4 (p. 2): *"What do you mean here — SLAM? More precision in language is needed"* (on "Conventional mobile mapping, which assumes continuous acquisition under motion and incremental online registration...")

Response 4 (p. 2): Yes — SLAM-based continuous mapping was meant, and it is now named: "...register incrementally online through simultaneous localization and mapping (SLAM)... SLAM-based continuous mobile mapping therefore does not

transfer directly to the stop-and-scan workflow addressed here.” (SLAM is now defined at this first use.)

Comments 5 (p. 2): *“what is the timeline for this?”* (on the contribution bullet “per-run offline registration of the autonomously acquired BLK360 scans...”)

Response 5 (p. 2): Section 6.7 now states when and how long the registration step takes: “Registration is a post-mission batch step; importing and aligning one run’s network took on the order of minutes on a desktop workstation, small compared with the acquisition time of the scans themselves.”

Comments 6 (p. 3): *“What are the restrictions of the path (dimensions)?”* (on “...the act of ‘observing’ is effectively free along the path.”)

Response 6 (p. 3): The physical restrictions on the robot’s paths are now stated in Section 5.1 with the actual safety-margin values: “All navigation is subject to the platform’s physical constraints: Nav2 plans on an inflated costmap, so every path and every scan candidate keeps the robot footprint (Figure 3b) clear of obstacles by the Nav2 costmap inflation margin (inflation radius 0.70 m with a 0.22 m robot radius in all reported runs).” The platform’s dimensions themselves are now given in the new annotated platform figure (see Comment 10): deck 71 × 49 cm at 31 cm height, wheelbase 28 cm, track 44 cm.

Comments 7 (p. 3): *“but the origin point is lower than a typical field of view; what are the challenges/drawbacks for this?”*

Response 7 (p. 3): A fair point that we now address in two places. Section 5.1 quantifies the vantage: “The deck places the scanner’s optical center roughly 0.5 m above the floor — lower than a typical 1.5–1.8 m tripod setup.” The Discussion (Section 7) analyzes the consequence: “the deck-mounted scanner observes from roughly 0.5 m above the floor, lower than a typical tripod setup, so low furniture and equipment cast longer horizontal shadows than in a manual survey of the same room. This makes occlusion-aware placement more important on a robotic platform, not less; but it also means the acquired clouds have a lower vantage than a conventional tripod survey would provide.”

Comments 8 (p. 3): *“again — this assumption can be easily challenged”* (on “each measurement requires a stationary dwell of minutes and registration is performed offline”)

Response 8 (p. 3): As in Comment 3, the claim is now scoped to the instrument class rather than stated universally: “Survey-grade terrestrial scanning, as practiced with the tripod-class instruments considered here, departs from the continuous-acquisition assumption... (Scanners that relax these constraints exist, but at accuracies below the survey grade targeted here.)”

Comments 9 (p. 9, Figure 1): *“colors too similar”*

Response 9 (p. 9, Figure 1): The architecture diagram has been redrawn (also at Reviewer 1’s request): the visibility-coverage block now uses a clearly darker fill and a heavy dark-blue border that separates it from the pale exploration boxes, the sequencer band uses a distinct warm tone, the survey product is green, and the redundant color legend was replaced by direct container labels. Labels were also enlarged roughly twofold.

Comments 10 (p. 10, Figure 2): *"This angle is difficult to understand the full scale and layout of the unit — it would useful to see an annotated set of shop drawings or a more axonometric view, with dimensions and/or scalar elements"*

Response 10 (p. 10, Figure 2): The platform figure has been replaced by a two-panel figure: (a) the photograph annotated with callouts for every component (Leica BLK360 G1, rigid scanner mount, YDLiDAR TG30 2D LiDAR, RGB-D camera, deck, chassis with battery/controllers/BLDC skid-steer drive); and (b) a dimensioned top-view schematic drawn from the platform's hardware specification: deck 71×49 cm at 31 cm height, wheelbase 28 cm, track 44 cm, wheels $\varnothing 21 \times 6$ cm, the 2D LiDAR 7 cm from the front deck edge, and the sensor positions marked, with the scanner optical-center height (≈ 0.5 m) noted.

Comments 11 (p. 11, Figure 3): *"dimensions? characteristics?"*

Response 11 (p. 11, Figure 3): The test room's characteristics are now given in the main text (Section 5.2) rather than the caption, including its extent: "Its mapped free space spans approximately 16×8.5 m, about 105 m^2 of free floor area on the reference map," together with the qualitative characteristics (open exhibition/staging space, flat reflective floor, display wall of monitors and product panels, freestanding equipment cases and furniture creating self-occlusion, floor markings for the survey reference layout).

Comments 12 (p. 13, Figure 5): *"explain colors"*

Response 12 (p. 13, Figure 5): The caption now defines the color convention, once for all placement figures: "Each accepted scan pose (marker) and its visibility polygon share one color, indexed by scan order (#1...#4); the grayscale background is the occupancy map (light gray free, black occupied, darker gray unknown) and the blue line is the robot path. The same color convention is used in all placement figures."